

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2021

EE 1213

(Electrical Circuits and Electronics)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

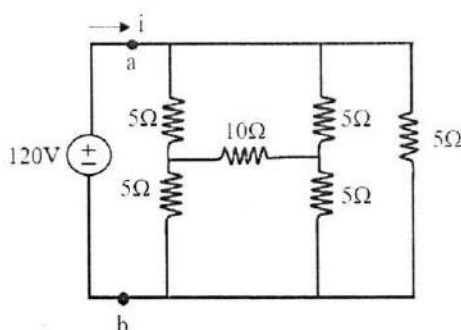
ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

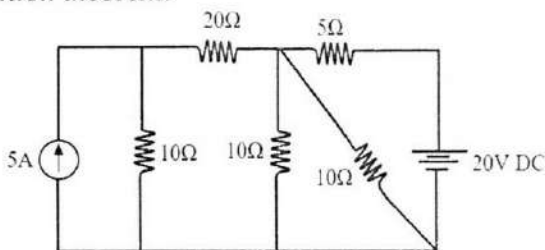
SECTION – A

- 1(a). Define emf and potential difference. What is meant by dependent source? What are the types of dependent source? 06

- 1(b). Obtain the equivalent resistance R_{ab} for the circuit shown in figure below and use it to find current i . 15



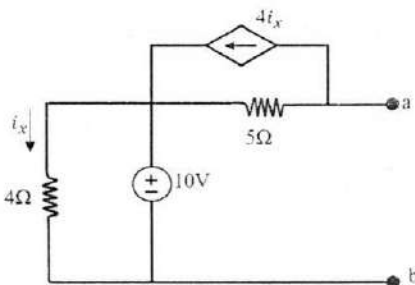
- 1(c). State superposition theorem. Find the voltages and currents of the circuit as shown in figure, using superposition theorem. 14



- 2(a). What is power factor? Draw the power triangle and show that capacitor can improve power factor. 10

- 2(b). State and explain maximum power transfer theorem. Show that, at maximum power transfer condition the efficiency is 50%. 12

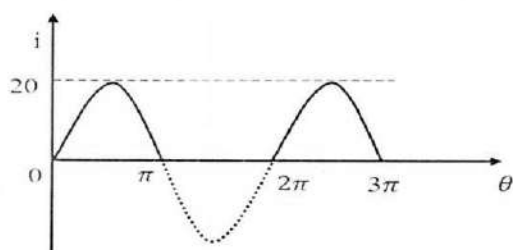
- 2(c). Using Norton's theorem, find R_N and I_N of the following circuit at terminal a-b. 13



- 3(a). Define i) average value, ii) RMS value, iii) Form factor. 09

- 3(b). Find the expression of i) Apparent power, ii) Real power and iii) Reactive power for $v = 20 \sin(\omega t - 30^\circ)$ and $i = 5 \cos(\omega t - 60^\circ)$. Also find the phase difference and power factor. 15

- 3(c). What is the significance of RMS value? Determine the form factor for the following waveform. 11



- 4(a). What is sinusoid? Using the phasor approach, determine the current $i(t)$ in a circuit described by the integro differential equation: 10

$$4i + 6 \int i dt + 5 \frac{di}{dt} = 30 \cos(2t + 60^\circ)$$

- 4(b). Derive the frequency resonance condition for parallel RLC circuit. Hence, derive the quality factor of the system. 15

- 4(c). Find the impedance expression for i) $\frac{(3+5i)(4-5i)}{\sqrt{(3+5i)}}$ ii) $\frac{(3+6i)(6-3i)}{\sqrt{(2(4+5i))}}$ 10

SECTION – B

- 5(a). Draw the energy band –diagram of i) intrinsic semiconductor, ii) p- type semiconductor and iii) extrinsic semiconductor. If a positive electric field of V volt is applied across them, what will be their band diagram? Draw them individually. 12

- 5(b). Why does electron mobility is higher than the hole mobility in semiconductor? Derive the mass-action law as , $n_i^2 = np$. 13

- 5(c). Given that the density of states related effective masses of electrons and holes in GaAs are 1.08 m_e and 0.06 m_e , respectively and electron and hole drift mobility in room temperature are 1350 and 450 $cm^2 v^{-1} s^{-1}$, respectively. Calculate the intrinsic concentration and intrinsic resistivity of GaAs 10

- 6(a). Write down the differences between Avalanche breakdown and Zener breakdown of semiconductor. Mention the working of Zener diode for voltage stabilization. 12

- 6(b). Explain the effect of temperature on carrier concentration. Draw the temperature dependence curve of electron concentration in an n-type semiconductor. 10

- 6(c). What is an ideal diode? From the diode equation, drawn the I-V characteristics curve of semiconductor diode and identify the significant region on the curve. 13

- 7(a). Suppose, you need to design a maximum power point tracker (MPPT) for solar-cell efficiency improvement. Which logic device do you suggest? Draw the circuit of the logic device. 10

- 7(b). Derive that, at no-bias condition in p-n junction, the depletion width of semiconductor junction is proportional to the square root of contact potential. 15

- 7(c). Let $f = \sum(5,6,14)$ and $f_1 = \pi(4,7,15)$, Determine f_2 such that $f = f_1 f_2'$. 10

- 8(a). What is latch? What are the features of edge triggering flip flop? 08

- 8(b). Design a counter with repeated binary sequence 0, 1, 2, 5, 6. Use T-flip flop. 12

- 8(c). Explain the minimization technique using Karnaugh map. Determine the logic gate for the following function: 15

$$F(A, B, C, D) = \pi(1,3,7) + \pi(13, 14, 15)$$

-End-

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2021

Hum 1213

(Technical English)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Transform the following sentences as directed. 14
- i) Though the tortoise was slow, it won the race. (Simple)
 - ii) Disobeying his parents, Rana went to cinema. (Complex)
 - iii) Since he did not feel well, he decided to stay at home. (Compound)
 - iv) The Reddison is more expensive than most other restaurant. (Positive)
 - v) Wasn't I sincere in my job. (Assertive)
 - vi) There is no smoke without a fire. (Affirmative)
 - vii) Riya always minds her lesson. (Negative)
- 1(b). Change the following words as asked in brackets and make sentence with the changed forms. 12
- Anxiety (into adj.); Illegitimacy (into adj.); Confident (into adj.); Supersession (into verb); Emancipate (into noun) ; Bath (into verb).
- 1(c). Make words with the following suffixes and prefixes and use them in sentences. 09
- il, mal....., y, non, de.....,al.
- 2(a). Make sentence with the following structures using the words given in brackets. 14
- (i) Subj. + Linking verb + Noun complement. (Appear as verb)
 - (ii) Subj. + Intransitive verb + Adv. of Manner. (Walk as verb)
 - (iii) Subj.+ Transitive verb + Object. (Write as verb)
 - (iv) That + subj.+ verb + Adv. of Manner + Verb + Adj. complement (Live and is as verb)
 - (v) Since + Subj. + Verb + Adv.of Manner, Subj. + Verb + Adv. (Study and do as verb)
 - (vi) Not only + subj. + but also + subj. + verb + Adv. (Attend as verb)
 - (vii) Subj.+ Verb + neither + Adj. complement + nor + Adj. complement. (is as verb)
- 2(b). Make use of the following words in sentence as asked in brackets. 12
- Above (as adv.); Above (as preposition); All (as adv.); Baby (as verb); Book (as verb); Cloud (as verb).
- 2(c). Write two synonyms for each of the following words and use the synonyms in sentence. 09
- i)Hope, ii)Perseverance, iii)Endeavor.

- 3(a). Frame Wh questions from the following answers: 14
- i) Sejan is studious boy.
 - ii) My uncle is a businessman.
 - iii) An accident occurred near our house.
 - iv) We found him in Room no. 203.
 - v) The essay sound revolutionary.
 - vi) She was sleeping deeply.
 - vii) The bridge is 40 feet wide.
- 3(b). Complete the following sentences with clauses as asked in brackets. 12
- (i) -----inspires us really. (Noun clause)
 - (ii) We call Ranu------. (Noun clause)
 - (iii) You will continue your studies------. (Adv. clause)
 - (iv) You will do the sum------. (Adv. clause of Manner)
 - (v) -----, I will feel compatible. (Adv. clause of condition)
 - (vi) Labib,-----, is a doctor (Adj. clause)
- 3(c). Express the following notions/functions in sentence. 09
- i) Sadness, ii) Determination, iii) Cheerfulness, iv) Honesty, v) Fear, vi) Greetings.
- 4(a). Correct the following sentences: 14
- i) Both Hera as well as Mera were there.
 - ii) I have no other friend but you.
 - iii) As he enterd the room, then he met me.
 - iv) He had scarcely gone than his friend arrived.
 - v) The prisoner was accused for muder.
 - vi) Until you work hard, you will improve.
 - vii) These all fishes are rotten.
- 4(b). Make use of the following modals in sentence as asked in brackets . 12
- i) May. (To allow somebody for something)
 - ii) May. (To express uncertainty)
 - iii) May. (To ask someone politely for something)
 - iv) Might. (To express uncertainty)
 - v) Must + have + past participle of verb. (To express logical deduction)
 - vi) Must. (To express certainty)
- 4(c). Make sentence with following idioms and phrases. 09
- i) Eat like a horse, ii) A hot potato, iii) Piece of cake, iv) Miss the change, v) Devil's advocate, vi) Hit the books.

SECTION – B

- 5(a). Read the passage and answer the following questions. 20

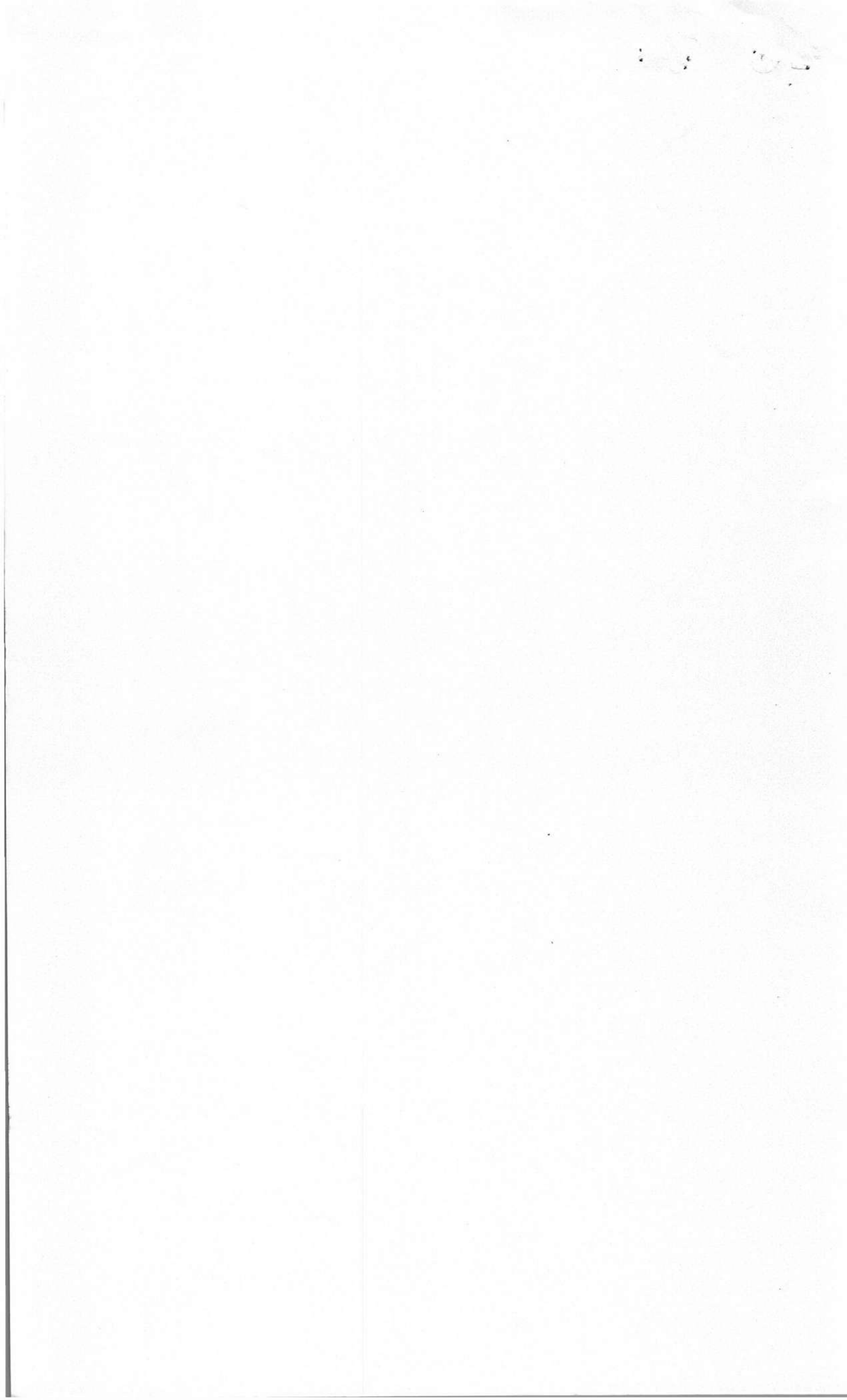
Teaching English to the native people of Bangladesh is not at all an easy task. In school, the traditional grammar-translation method is followed so the pupils only develop their reading and writing skills but they cannot speak or understand English by listening. So a direct method, which gives stress to listening and speaking, has been developed. The teacher has to draw pictures, act to create a situation and has to show models or real object. It is creating a bond between experience and expression. It is time consuming because the teacher has to create a situation of English speaking environment, which is a tiresome task. We also know that every language has a definite structural pattern and in essential English, there are 282 structures. Following the structure, the teacher gives several examples, then goes for oral drill and invites more examples from the students. This process also lays emphasis on some written work like making sentences using the substitution tables. But when the oral drill goes on, the class becomes noisy and sometimes limited to a particular pattern. It is mechanical and monotonous. So, communicate approach has evolved, where interaction between the teacher and the students take place as two-way process. Audio-lingual approach also has evolved as modern concept where there is a scope to follow the English speaker to improve listening and speaking skills as well as pronunciation through the cassettes. Teacher should know all the methods. The blending of approaches will be the teachers own creativity.

Questions :

- (i) Why is teaching English to native people of Bangladesh not an easy task?
- (ii) Which of methods can be suitable for university students? Why?
- (iii) This semester you gave stress on listening and speaking. How far do you think you have improved?
- (iv) Is language learning a monotonous activity or is it only applicable for English language learning in the context of Bangladesh?

- 5(b). Make a précis of the above passage (Question 5.(a)) with a title. 15
- 6(a). Amplify the idea- “ If winter comes can spring be far behind” 20
- 6(b). Write a paragraph on the qualities of a good student. 15
- 7(a). Prepare your CV for post of lecturer in the Department of Energy Science and Engineering and apply for the post. 20
- 7(b). Suppose your department has arranged a program to discuss how to become successful. Write a memo to inform the students of ESE about the program. 15
8. Write a free composition on any of the followings. 35
- i) A life in deeds.
 - ii) Honesty: A rich legacy.

-End-



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering
B. Sc. Engineering 1st Year 2nd Term Examination, 2021
ESE 1205

(Thermodynamics for Energy Engineering)

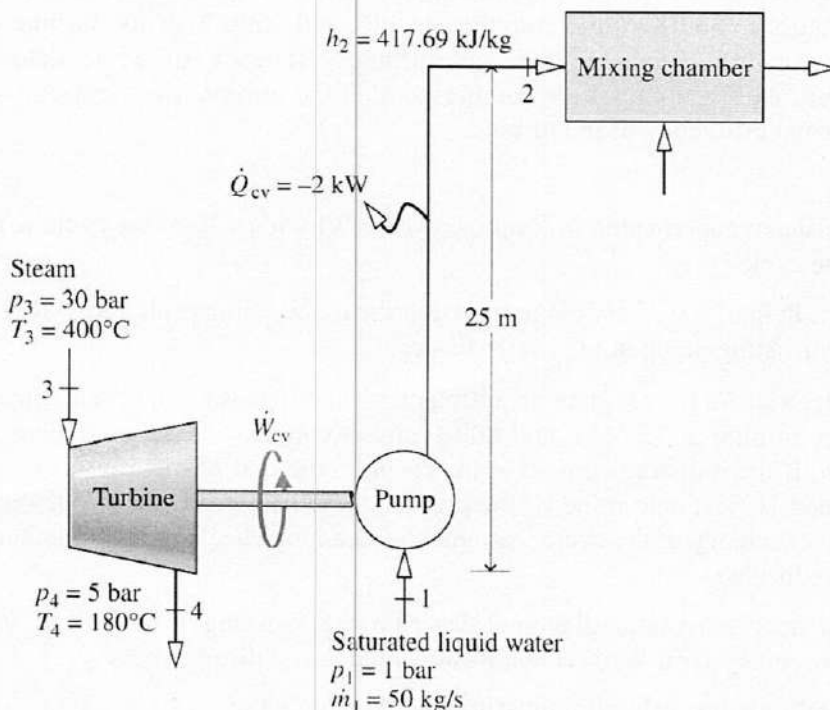
Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Why heat and work are path function but thermodynamic properties are point function? Describe. 10
- 1(b). Briefly explain the Stored and Transient form of energy with the help of schematic diagram. 10
- 1(c). Define continuum? Write down the Gibb's phase rule for non reacting system and mention the meaning of each term. 05
- 1(d). 0.45 kg of air in a piston-cylinder assembly undergoes a thermodynamic cycle consisting of three following process: 10
Process 1–2: constant specific volume
Process 2–3: constant temperature expansion
Process 3–1: constant pressure compression
At state 1, the temperature is 300 K, and the pressure is 101.325 kPa. At state 2, the pressure is 200 kPa. Employing the ideal gas equation of state, (ii) sketch the cycle on p - v coordinates, (ii) determine the temperature at state 2, and (iii) determine the specific volume at state 3 in m^3/kg .
- 2(a). State zeroth and first law of thermodynamics. Write down the mass and energy rate balance equation for a control volume and mention what each term in the equation mean. 10
- 2(b). What are the differences between steady state and transient analysis of a system? Explain with example. 05
- 2(c). Following figure shows a turbine drives pump that provides water to a mixing chamber located at 25 m height than the pump. Steady state operating data for the turbine and pump are labeled on the figure. Heat transfer from the water to its surroundings occurs at a rate of 2 kW. For the turbine, heat transfer with the surroundings and potential energy effects are negligible. Kinetic energy effects at all numbered states can be ignored. Determine: 20
(i) the power required by the pump in kW to supply the water to the inlet of the mixing chamber.
(ii) the mass flow rate of steam in kg/s that flows through the turbine.



- 3(a). State Clausius, Kelvin-Plank and entropy statement of second law of Thermodynamics. 05
- 3(b). State Clausius inequality and prove that, $ds = \frac{\delta Q}{T} \Big|_{int rev}$. 10
- Also write down the entropy rate balance equation for a control volume and mention what each term in the equation mean.
- 3(c). Steam enters an adiabatic turbine at 6 MPa, 600 °C, 80 m/s, and leaves at 50 MPa, 100 °C, 140 m/s. If the power output of the turbine is 5 MW, determine (i) mass flow rate of the steam, (ii) the reversible power output, (iii) the isentropic efficiency of the turbine, and (iv) the second-law efficiency of the turbine. 20
- Assuming the surroundings to be at 25 °C.
- 4(a). How does useful work differ from actual work? Define entropy and exergy. Show that exergy of a control mass system is given by: 12
- $X = (U - U_0) + P_0(V - V_0) - T_0(S - S_0) + KE + PE$, where the symbols have their usual meaning.
- 4(b). Write down the exergy balance equation for an open system and mention its each term with their meaning. 05
- 4(c). When a Real gas can be treated as Ideal gas? Also derive the first "Tds" and second "Tds" equation of a system. 08
- 4(d). Predict the pressure of nitrogen gas at temperature, $T = 175$ K and density, $\rho = 0.00375$ m³/kg on the basis of (i) the ideal gas equation of state, and (ii) the Van der Waals equation of state. Compare the values obtained to the experimentally determined value of 10 MPa. For nitrogen, the critical pressure and temperature are 3.39 MPa and 126.2 K, respectively. 10

SECTION – B

- 5(a). Write down the air standard assumptions of an actual cycle. What are the differences between vapor cycle and gas power cycle? 08
- 5(b). With the help of p-v and T-s diagram, show that the efficiency of Otto cycle is greater than the efficiency of Dual and Diesel cycle for same compression ratio and heat rejection. Why this happens? Explain. 15
- 5(c). A gas turbine power plant is operating on Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Assuming a compressor efficiency of 80% and turbine efficiency of 85%, determine (i) the gas temperature at the exits of the compressor and the turbine, (ii) the back work ratio, and (iii) the thermal efficiency of the turbine. 12
- 6(a). Why is steam superheated in Rankine cycle? Why ideal Rankine cycle is modified to reheat Rankine cycle? 10
- 6(b). With the help of necessary diagrams, describe the working principle of Regenerative Rankine cycle consisting an open feedwater heater. 10
- 6(c). Consider a steam power plant operating on ideal reheat Rankine cycle. Steam enters the high-pressure turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. If the moisture content in the steam at the exit of the low-pressure turbine is not to be exceed 10.4%, determine (i) the pressure at which steam should be reheated, and (ii) the thermal efficiency of the cycle. Assume the steam is reheated to the inlet temperature of high-pressure turbine. 15
- 7(a). With a neat schematic diagram, describe the working principle of vapor compression refrigeration system. Write down its advantages and disadvantages. 12
- 7(b). State Dalton's law of partial pressure and Amagat's law. 04

- 7(c). A gas mixture has composition on a mole basis of 60% O₂ and 40% CO₂ at a pressure of 300 kPa and a temperature of 20°C. Calculate (i) mass fraction of each constituent, (ii) equivalent molecular weight of the mixture, (iii) equivalent gas constant of the mixture, (iv) partial pressure and partial volume, and (v) volume, density, C_p and C_v of the mixture. Assume γ for CO₂ and O₂ are 1.29 and 1.40, respectively. 13
- 7(d). Write short note on the following terms: 06
(i) Gas mixture, (ii) Gravimetric analysis, and (iii) Partial pressure of a gas mixture.
- 8(a). With the help of psychrometric chart, describe the change of specific humidity and relative humidity of moist air during cooling and dehumidification processes. 10
- 8(b). Write short note on the following terms: 10
(i) Psychrometry, (ii) Relative humidity, (iii) Moist air, (iv) Wet bulb depression, and (v) Dew point temperature.
- 8(c). Atmospheric air at 1.0132 bar has a dry bulb temperature of 30°C and wet bulb temperature of 27°C. Compute the: (i) partial pressure of water vapor, (ii) specific humidity, (iii) dew point temperature, (iv) relative humidity, (v) degree of saturation, (vi) density of air, and (vii) enthalpy. 15

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2021

Math 1213

(Differential Equation and Co-ordinate Geometry)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Find the cylindrical and spherical polar co-ordinates for the point $(-4, 2, 1)$. 12
- 1(b). Reduce the equation $4x^2 - 4xy + y^2 - 8x - y + 6 = 0$, to the standard form. Find its any two properties. 13
- 1(c). Find the transformed equation of $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ when the origin is shifted to $(2, -3)$. 10
- 2(a). Define direction cosine and direction ratio of a line. Also find the relation between them. 10
- 2(b). Transfer the equation $10x^2 + 15xy + 5y^2 - 10x - 20y = 10$ to rectangular axis through the point $(2, -1)$ and inclined at an angle $\tan^{-1}(3/4)$ 15
- 2(c). Is that the four points $(-1, 2, 3)$, $(6, 1, 3)$, $(3, 2, 4)$, $(7, 0, -1)$ are coplaner? 10
- 3(a). Define dot and cross product with example. Find the angle which the vector $3\bar{i} - 6\bar{j} + 2\bar{k}$ makes with the co-ordinate axes. 10
- 3(b). Find the vector equation of the line that passes through the point $(0, 0, -2)$ and $(5, 3, 8)$. Also find the parametric equation of this line. 12
- 3(c). Find the length of the shortest distance between the lines: 13
- $$\frac{x-1}{2} = \frac{y+3}{1} = \frac{z}{3} \text{ and } \frac{x+1}{3} = \frac{y}{2} = \frac{z-1}{1}$$
- Also find the equation of the line.
- 4(a). Find the vector equation of the plane which passes through the origin and is parallel to the vector $\bar{a}$ and $\bar{b}$. 10
- 4(b). Find the equation of the plane through $(3, 2, -1)$ and perpendicular to each of the planes $\bar{r} \cdot (2\bar{i} - \bar{j} + 2\bar{k}) = 0$ and $\bar{r} \cdot (\bar{i} + 3\bar{j} - 5\bar{k}) + 1 = 0$. 12
- 4(c). A plane meet the co-ordinate axes in A, B, C such that the centroid of the triangle ABC is at the point (α, β, γ) . Find the equation of the plane. 13

SECTION – B

- 5(a). Define order, degree, general solution and particular solution of a differential equation. Also write two characteristic properties of linear ordinary differential equation. 10
- 5(b). Write the general form of an n-th order initial value problem (IVP). Find the value of C_1 and C_2 such that $\phi(x) = C_1 e^{2x} + C_2 e^{-x}$ is a solution to the IVP $y'' - y' - 2y = 0$ subject to $y(0) = 2$ and $y'(0) = -3$. 10
- 5(c). Find the general solution to the ODE: 15

$$\frac{dy}{dx} = -\frac{3y - 7x + 7}{7y - 3x + 3}$$

6(a). Solve: $\frac{dy}{dx} = (4x + y + 1)^2$; $y(0) = 1$ 12

6(b). Solve: $\frac{d^2y}{dx^2} + y = x \cos(x)$ 10

6(c). Find the particular solution to the IVP: 13

$$2 \sin(y^2)dx + xy \cos(y^2)dy = 0; \quad y(2) = \sqrt{\frac{\pi}{2}}$$

Also define homogeneous function and total derivative of $z = f(x, y)$.

7(a). Find the solution of $y'' = y$ which passes through the point $(\ln 2, \frac{3}{4})$. 10

7(b). Solve the following by the method of variation of parameter: 13

$$\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^{2x}}{1 + e^{2x}}$$

7(c). Find the steady-state charge and the steady-state current in an LRC series circuit when $L = 1 \text{ h}$, $R = 2 \Omega$, $C = 0.25 \text{ f}$ and $E(t) = 50 \cos t$. Is the circuit overdamped? 12

8(a). Find the Laplace transform of $f(t) = e^t t \sin^2 t$ and $g(t)$ where $g(t)$ has the following graph. 11

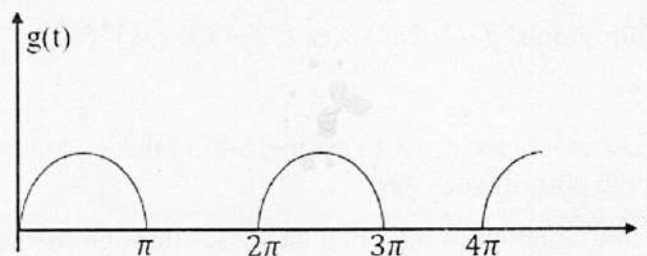


Figure: half-wave rectification of $\sin t$

8(b). Find the inverse Laplace transform of $\frac{s+\pi}{s^2+\pi^2} e^{-s}$ and $\frac{1}{(s^2+c_0^2)^2}$ 12

8(c). Solve the ODE using Laplace transform 12

$$y'' + y = f(t), \quad y(0) = 0, \quad y'(0) = 0 \text{ where}$$

$$f(t) = \begin{cases} 1, & 0 \leq t < \pi/2 \\ \sin t, & t \geq \pi/2 \end{cases}$$

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 1st Year 2nd Term Examination, 2021

Course No Ch. 1213

(Chemistry II)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

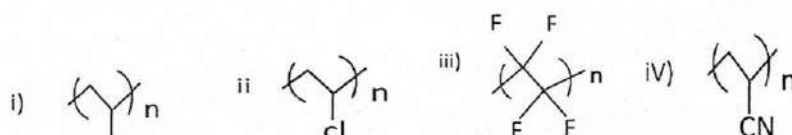
- 1(a). Define order and molecularity of a reaction? 06
- 1(b). The water "flea" *Daphnia* performs a constant number of heart beats and then dies. On average, the flea lives twice as long at 15°C as at 25°C. Estimate the activation energy for the reaction that controls the rate of its heart beat? 07
- 1(c). Deduce the expression for the rate constant of the following type of reaction 10
 $3A \rightarrow \text{Products}$
- 1(d). Explain the simple collision theory of the reaction rate. Also give the reasons for its failure. 12
- 2(a). What are meant by chemiluminescence, Phosphorescence and fluorescence? 09
- 2(b). Distinguish between photochemical and thermal reactions. 07
- 2(c). What is the Beer's law and Beer-Lambert's law? Derive the Beer-Lambert law. Mention the limitations of this law. 09
- 2(d). How would you explain very high and very low quantum efficiencies of some photochemical reactions? 10
- 3(a). Explain the terms with suitable examples: i) Desorption, ii) Sorption and iii) Occlusion. 09
- 3(b). Distinguish between physical and chemical adsorption. What are adsorption isobars? 08
- 3(c). Derive Langmuir isotherm equation. How does this isotherm interpret the kinetics of unimolecular reactions catalyzed by the solid surfaces? 13
- 3(d). Discuss the role of ion-exchanger in water softening technique. 05
- 4(a). Explain the followings with example: 09
i) Catalytic promoters, ii) Autocatalysis, iii) Catalytic poisons.
- 4(b). What are the characteristics of catalytic reaction? 07
- 4(c). Derive Michaelis-Menten equation for enzyme catalyzed reaction. 11
- 4(d). "Adsorption is an exothermic process" explain. 08

SECTION – B

- 5(a). What are the distinguishing features of sp^3 and sp^2 hybrid orbitals? Explain with suitable examples. 10
- 5(b). Two compounds have the same molecular formula, $C_2H_2Br_2$. One has a dipole moment; the other does not. What is the identity of the two compounds? Relate the dipole moments to the structure and describe the argument you use? 09
- 5(c). Why is the rotation about a double bond restricted? 08
- 5(d). Define specific rotation? How is the observed rotation related to specific rotation? 08

6(a). Write down the monomers of the following polymers:

08



6(b). Write down the major steps of addition polymerization?

10

6(c). What is electrically conducting polymers? How does I_2 doping increase the conducting of polyacetylene?

09

6(d). Assume E & S denote enzyme and substrate. The reaction between E&S follows Michaelis-Menten Kinetics. Draw the reaction rate vs substrate concentration graph for this reaction. Identify the first order and zero-order kinetics region in the graph. Show the maximum reaction rate and Michaelis-Menten constant in the graph. Describe how did you determine these regions.

08

7(a). Distinguish between corrosion and erosion with suitable examples.

05

7(b). Using the Nernst equation, explain how the corrosion of iron would be affected by the p^H of water? Using the same formula explain what other parameter can influence this corrosion?

13

7(c). Steel corrodes in an aqueous solution, the corrosion current is $0.1A/cm^2$. Calculate the rate of weight loss per unit area in units of mdd ($mgdm^{-2}day^{-1}$)?

10

7(d). Mention three strategies that can be used to prevent corrosion?

07

8(a). Write down the necessary conditions for a molecule to absorb infrared light?

07

8(b). The frequency of light that a molecule will absorb is given by:

07

$$\nu = \frac{1}{2\pi} (k/\mu)^{\frac{1}{2}}; \nu = \text{frequency in } cm^{-1}, k = \text{force constant in N/cm,}$$

$\mu = \text{reduced mass in Kg.}$

Using this formula, explain why $C^{-2}D$ stretching frequency will be reduced compared to $C^{-1}H$ stretching?

8(c). Draw the different vibrational modes for a three atomic molecule?

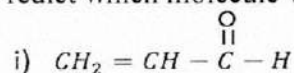
08

8(d). Describe why symmetric stretch of CO_2 is not IR active?

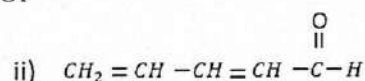
06

8(e). Predict which molecule will absorb less energy in the UV-visible region and mention the reasons

07



Or



-End-